

Credit Co-Pilot

Agentic Loan Underwriting Assistant

High-Level Design | Tech Stack | System Architecture

Claude Opus 4.7 • LangGraph • AWS EKS • Temporal • Postgres • Kafka
OPA • SageMaker • pgvector • Feast • MLflow • Datadog

Executive Summary

Credit Co-Pilot is an agentic AI system that assists human underwriters by automating data gathering, analysis, and risk-narrative drafting for loan applications. The agent ingests an application packet (PDFs, bank statements, tax returns, bureau pulls), reasons over policy rules, computes risk metrics, and produces an auditable underwriting memo with a recommended decision (approve / decline / refer-to-human).

Design pillars

- **Human-in-the-loop** — the agent recommends, the underwriter decides.
- **Auditability** — every decision traces to source documents and policy clauses.
- **Compliance-first** — ECOA, FCRA, GLBA, GDPR, SR 11-7 baked in.
- **Modular** — swappable models, tools, and data sources.

Headline KPIs (target)

Cycle time / file	–30% Phase 1	–55% Phase 2
Memo factuality (golden set)	≥ 95%	≥ 98%
Underwriter override rate	< 20%	< 10%
Default rate vs. baseline	≤ baseline	≤ baseline

System Architecture

End-to-end view of components, data flow, and the AI/ML, knowledge, decisioning, and platform planes. Tech logos are used to indicate the canonical implementation choice for each role.

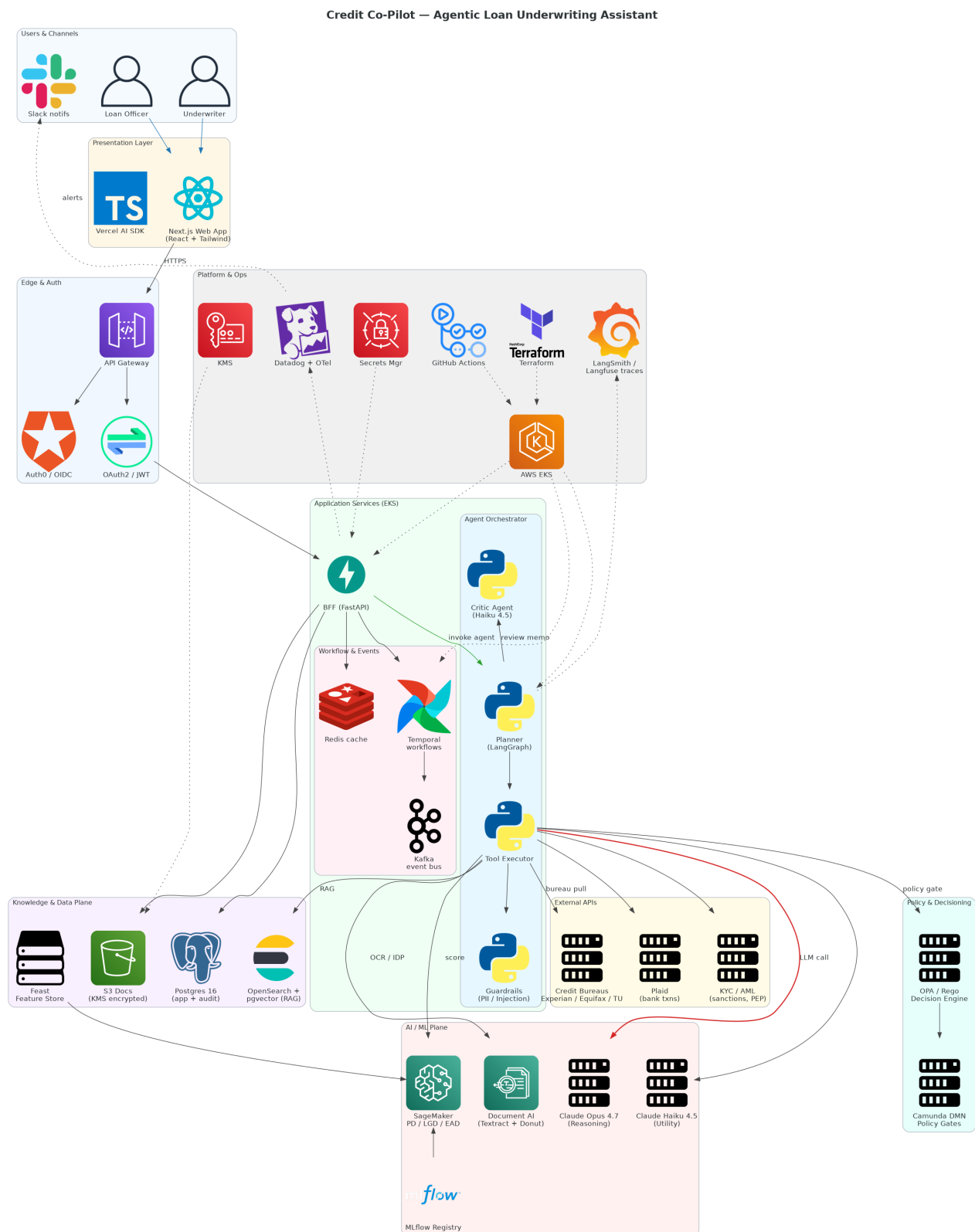


Figure 1. Credit Co-Pilot reference architecture. Solid arrows indicate runtime data flow; dotted arrows indicate platform/observability dependencies.

Core Components

Agent Orchestrator

LangGraph state machine on the Anthropic SDK. **Claude Opus 4.7** for planning and memo drafting; **Claude Haiku 4.5** for cheap classification, extraction, and routing. Prompt caching pins the policy manual + underwriting playbook (~70% cost reduction). Tool-use loop runs until the terminal `submit_memo` tool is invoked. Temperature 0 for numeric extraction and policy checks; higher only for narrative.

Document Intelligence (IDP)

AWS Textract / Azure Document Intelligence for OCR + tables; LayoutLMv3 / Donut for structured fields (W-2, 1040, paystub, bank statement). Pydantic schema validation; fraud signals from EXIF, font consistency, MRZ checksums.

Retrieval (RAG)

pgvector (small scale) or OpenSearch / Pinecone (scale). Voyage-3 or text-embedding-3-large embeddings. Hybrid BM25 + dense + reranker (Cohere Rerank or Haiku-as-judge). Corpus = policy, product matrices, regulatory bulletins, de-identified prior memos.

Risk & Scoring

PD/LGD/EAD models (XGBoost / LightGBM, survival models) served via MLflow + SageMaker. Feast feature store. SHAP values surfaced in the memo for explainability.

Decision Engine

Open Policy Agent (Rego) or Camunda DMN for hard policy gates — LTV caps, OFAC, jurisdiction, age. Agent proposes; rules engine adjudicates.

Guardrails

PII redaction, prompt-injection filter on borrower text, structured-output validator, policy-violation classifier (prohibited basis under ECOA), capability-token-gated write tools, and a Haiku critic agent that reviews the memo for hallucinated citations and disparate-impact red flags before submission.

Human-in-the-Loop UX

Underwriter sees structured memo with hover-to-source citations, risk-score breakdown, agent tool-call trace, and one-click overrides. Feedback feeds offline eval and (with compliance review) RLHF/DPO datasets.

Tech Stack

Layer	Choice
LLM	Claude Opus 4.7 (reasoning) + Haiku 4.5 (utility) via Anthropic API
Agent framework	LangGraph + Anthropic SDK tool use
Backend	Python 3.12, FastAPI, Pydantic v2
Frontend	Next.js 14, React, Tailwind, shadcn/ui, Vercel AI SDK
Auth / RBAC	Auth0 / Okta, OAuth2 + OIDC, Cerbos
Storage	Postgres 16, S3 (KMS), pgvector / OpenSearch
Event bus	Kafka / Kinesis
Workflow	Temporal (long-running async tasks)
ML serving	MLflow + KServe / SageMaker
Feature store	Feast
Rules	OPA (Rego) or Camunda DMN
Observability	OpenTelemetry, Datadog, LangSmith / Langfuse
Eval	Promptfoo + custom golden set; offline shadow scoring
Infra	AWS EKS, Terraform, GitHub Actions
Secrets	AWS Secrets Manager + KMS

Agent Tool Surface

fetch_credit_bureau — Experian / Equifax / TransUnion pulls.

extract_document — IDP wrapper, schema-validated.

categorize_transactions — Plaid + transaction classifier.

compute_ratios — Deterministic DTI, DSCR, LTV.

lookup_policy — RAG over the policy manual.

score_pd — Calls the PD ML model service.

run_kyc_aml — Sanctions, PEP, adverse media.

request_stipulation — Asks the borrower for a missing doc.

submit_memo — Terminal action — writes to system of record.

Happy-Path Sequence

1. Loan officer uploads packet → S3; event on Kafka.
2. Temporal workflow: IDP extracts each doc; structured fields validated.
3. Bureau + KYC tools called in parallel; results cached.
4. Agent run starts: planner builds task list — verify identity → assemble income → compute ratios → policy check → score → draft memo.
5. Agent executes tools; every call logged with I/O and cost.
6. Decision engine evaluates hard gates; agent receives verdict.
7. Critic agent reviews draft; on pass, memo persisted and shown to underwriter.
8. Underwriter approves / edits / declines; downstream booking system notified.

Compliance, Risk & Governance

- **SR 11-7 / OCC 2011-12** — model inventory, independent validation, ongoing PSI/KS/AUC monitoring.
- **Fair lending** — disparate-impact testing; prohibited-basis filter on inputs.
- **FCRA §615** — adverse-action reason codes attached to every decline.
- **Data** — PII tokenized at rest; access logged; per-jurisdiction retention.
- **Reproducibility** — prompts, model versions, and tool versions pinned per agent run; deterministic replay.
- **Change management** — prompt + policy changes flow through PR review, eval-gated CI, and staged rollout (shadow → canary → 100%).

Phased Rollout

Phase	Scope	Exit criteria
0. Spike	Personal loan, 1 IDP doc type, read-only memo draft	Memo factuality ≥ 95%
1. Assistive	Full doc set, agent drafts, human decides	Underwriter time / file –30%
2. Co-pilot	Auto-approve low-risk band under thresholds	Default ≤ baseline, override < 10%
3. Expand	SMB / mortgage, multi-language	Per-product eval gates

Open Questions

- Build vs. buy for IDP (Textextract vs. Hyperscience vs. in-house Donut)?
- On-prem deployment requirement for any jurisdictions? If yes, evaluate Claude on Bedrock + private VPC.
- Tolerance for fully-automated decisions in any band, or always human-in-the-loop at launch?
- Source of truth for policy: PDF manual vs. structured DSL?